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Key Components of Llamafile

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Developing Portable CLI with Cosmopolitan

Running Llamafile

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Key Terms

Header Files - Declare functions and data types from the C standard library. Examples: stdio.h, string.h

main() Function - The entry point where execution of a C program begins.

printf() - Prints formatted output to the standard output stream.

Return Value - Value returned by main() to indicate success (0) or failure (non-zero).

This example shows these concepts

```
1 // Prints array of fruits
2 #include <stdio.h>
3
4 int main() {
5     char* fruits[] = {"apple", "banana", "cherry"};
6     int i;
7
8     for(i = 0; i < 3; i++){
9         printf("Fruit: %s\n", fruits[i]);
10    }
11
12    return 0;
13 }
```

Run

Reset

Fruit: apple
Fruit: banana
Fruit: cherry

Arguments (argc, argv[]) - Command line arguments passed when executing the program.

This example shows accepting input from the command line

```
1 // Get fruit from command line
2 #include <stdio.h>
3
4 int main(int argc, char *argv[]) {
5     if(argc > 1){
6         printf("You passed fruit: %s\n", argv[1]);
7     } else {
8         printf("Pass a fruit!\n");
9     }
10
11    return 0;
12 }
```

Run

Reset

Pass a fruit!

Cosmopolitan Libc

- Build once, run anywhere portability for C like Java
- Uses custom cosmo compiler instead of gcc
- Can output portable binaries for Linux/Windows/MacOS

Debugging Options with Cosmopolitan Libc

- Print system call traces with --strace
- Low level logging with --ftrace
- Redirect output to file

Integration with LlamaFile

- Cosmopolitan is core of LlamaFile for large language models
- Appreciate elegance by understanding lower level libs

Go to next item

✓ Completed

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